

COMP11107 - BUSINESS DATA COMMUNICATION & NETWORKS

Muhammad Taimoor B01828373 Paisley Campus

MSc IT Project Management

Why we need Ethical and Professional Consideration?

Designing a network for UWS Buildings 1 & 2 serves 500+ students, 20 admin staff and 20 academics. Ethical responsibilities are legally and professionally required under UK GDPR and BCS Code of Conduct.

- 1. Data Privacy:** Personal data of students and staff, academic records and login information must be kept private. The UK GDPR requires minimal data collection and secure storage of data. The network should be secured with encryption (WPA3, TLS) and access restrictions.
- 2. Data Transparency:** The user has to be notified about the data being collected, its usage and access. UWS must have clear policies in place regarding the use of the network and make sure that audit trails are only accessible to authorised users.
- 3. Right to Data Control:** Allowed to request access to, and deletion of, personal data (Article 17 GDPR). Role Based Access Control (RBAC) is provided to control access to information so that staff and/or students can only access information that is relevant to them.

Ethical Practices for Network Design

All hardware that is proposed for UWS Buildings 1 & 2 should be sourced ethically, secure by design, accessible and environmentally responsible:

- 1. Routers & Switches (Cisco/Juniper):** Must be purchased through certified sources, no conflict minerals. All 500+ users have to be protected by firmwares that are regularly updated to patch vulnerabilities.
- 2. The hardware firewalls:** Should not be configured to carry out deep packet inspection of personal content without user consent, in accordance with the privacy rights of the UK GDPR.
- 3. Wireless Access Points (WAPs):** Accompliment to the RF emission standards (ICNIRP). Disabled users to have the same access to coverage as everyone else in accordance with Equality Act 2010.
- 4. Cat6A Cabling & Fibre Backbone:** Health & Safety at Work Act 1974 - Safe installation required. Old cable is safely disposed of in an ethical way, preventing toxic landfill.
- 5. Servers (HP ProLiant):** They have to keep data in UK/EEA to meet GDPR requirements in the UK. The use of Energy Star servers helps to save the environment, which is an institutional moral commitment.

Ethical Practices for Network Design

Any software used throughout Buildings 1 & 2 should respect user rights and be legally licensed, and must not infringe on academic integrity:

- 1. VPN (Cisco AnyConnect):** Secures remote access by employees and students. Must not allow unauthorised watching. Follows the UK Investigatory Powers Act 2016 — records only as stipulated by legislation.
- 2. Firewall Software (pfSense/Cisco ASA):** Should not be used to filter out legitimate academic research. Ethical content filtering should not be a disproportionate limitation of freedoms of students.
- 3. VLAN Management:** Segments staff/student traffic. No data leakage is allowed across VLANs. Applies the "least privilege" principle to ensure that student access to admin data is protected.
- 4. Network Monitoring (SolarWinds/PRTG):** Cannot store personal browsing information. Anonymisation should be performed on usage analytics. Monitoring of activity will be for security reasons and all staff must be informed of this (transparent monitoring policy).
- 5. Microsoft Active Directory & Azure/M365:** RBAC – users only have access to the required systems. Compliance with the GDPR in the UK and storing data in UK datacentres. All must have access features (screen reader/ captioning).

Ethical Challenges

1. **There is a difference between surveillance and security:** Network monitoring tools (PRTG, SolarWinds) can gather fine-grain user activity data. The challenge with securing the network against threats and attacking student/staff privacy is an ethical one.
2. **Digital Divide & Accessibility:** There are some students who may not have personal devices to access. Where the school has a BYOD policy, students without devices are disadvantaged, which is a concern for equity and accessibility in line with the Equality Act 2010.
3. **Data Retention & Right to Erasure:** Network logs and stored credentials could be kept for longer than required. There is a conflict with the right to erasure under UK GDPR Article 17, which conflicts with IT teams' requirements for audit trails and compliance records.
4. **Consent & Acceptable Use:** Monitoring of network is subject to informed consent by students and staff. Implementing acceptable use policies without clearly communicating them is not in keeping with the principles of transparency.
5. **Vendor Ethics & Supply Chain:** University's hardware purchase from unethical suppliers (use of conflict minerals or unfair labour) is considered to be harming the university. The BCS Code of Conduct requires ethical procurement to be a professional responsibility.

Proposed Remedies

1. **Anonymised network monitoring.** Establish a Network Monitoring Policy. Log metadata only (not content). Ensure Staff / Student Consent through a UWS Acceptable Use Policy (AUP).
2. **Digital Divide:** Make device loan schemes available within the campus and ensure proper wired/wireless connectivity in all Labs and Common rooms. Provide a minimum of 4 Computer Labs per floor in both buildings as proposed in group CW.
3. **Define and implement a Data Retention** Schedule (max 12 months for Logs). Have automated deletion procedures in place. Appoint a Data Protection Officer (DPO) as required by UK GDPR Article 37.
4. **Consent & Acceptable Use**
All users must accept an AUP upon first use. The AUP should explicitly define monitoring scope, penalties and user rights, which is reviewed annually by IT governance.
5. **Vendor Ethics**
Adopt an Ethical Procurement Policy. Evaluate all suppliers in accordance with BCS Code of Conduct. Choose vendors that are certified to ISO 14001 (environmental) and SA8000 (social accountability), like Cisco and HP.

Key References

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